

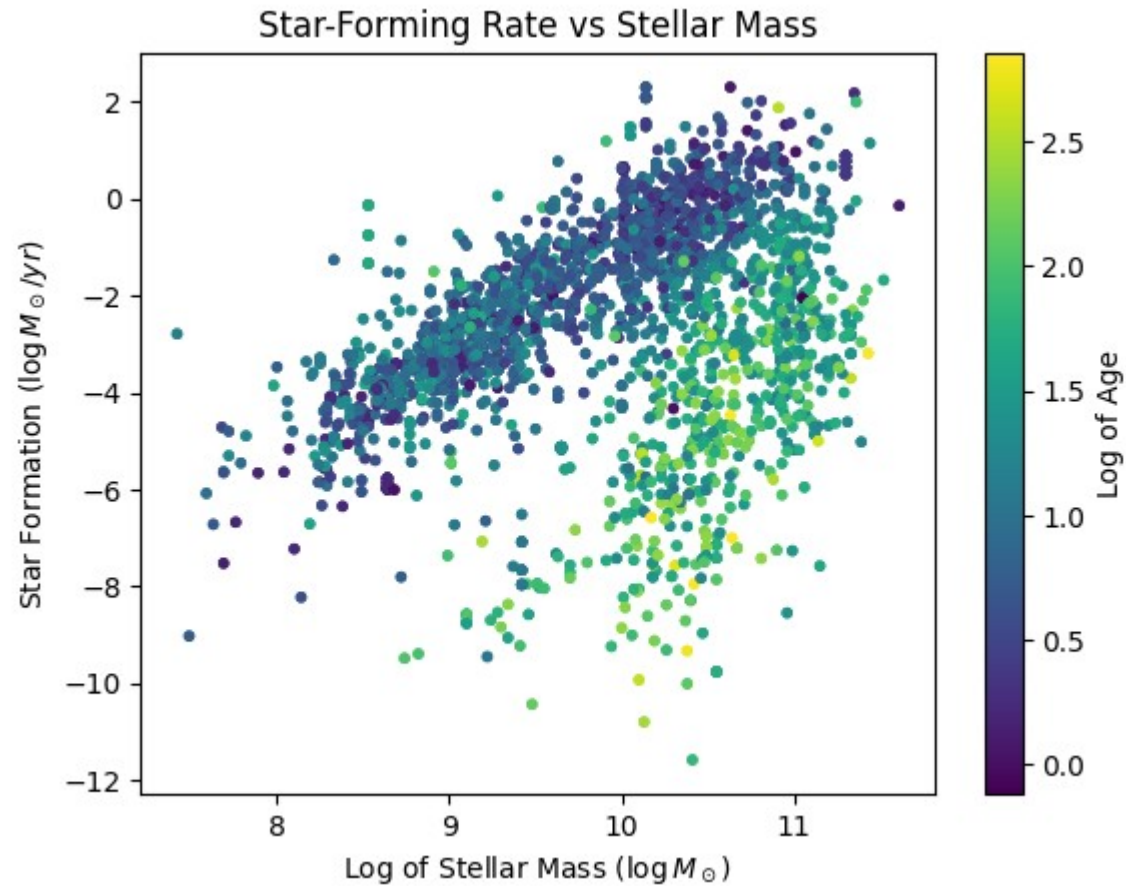
PHYS3116 Tutorial 4:

Star forming and quiescent galaxies

SAMI Survey

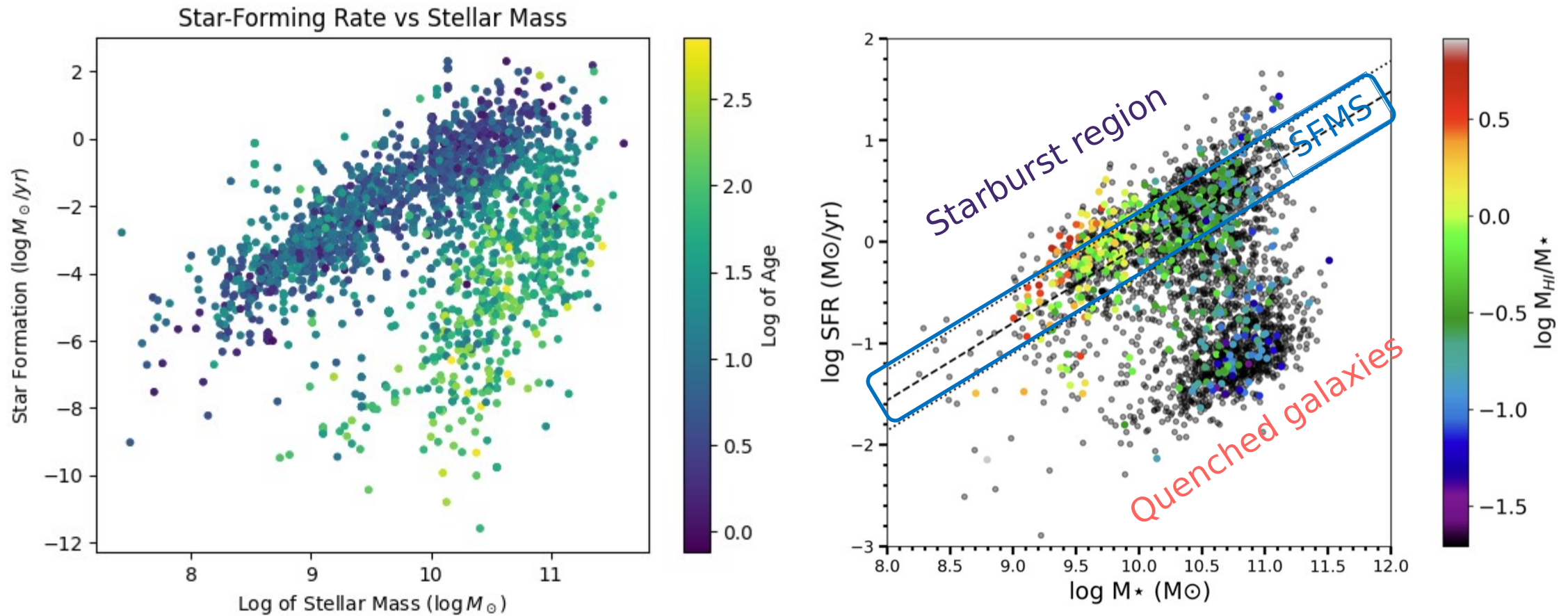
Star Formation Rate, Mass, Age, Spectra, Linear Regression

What can this graph tell us?



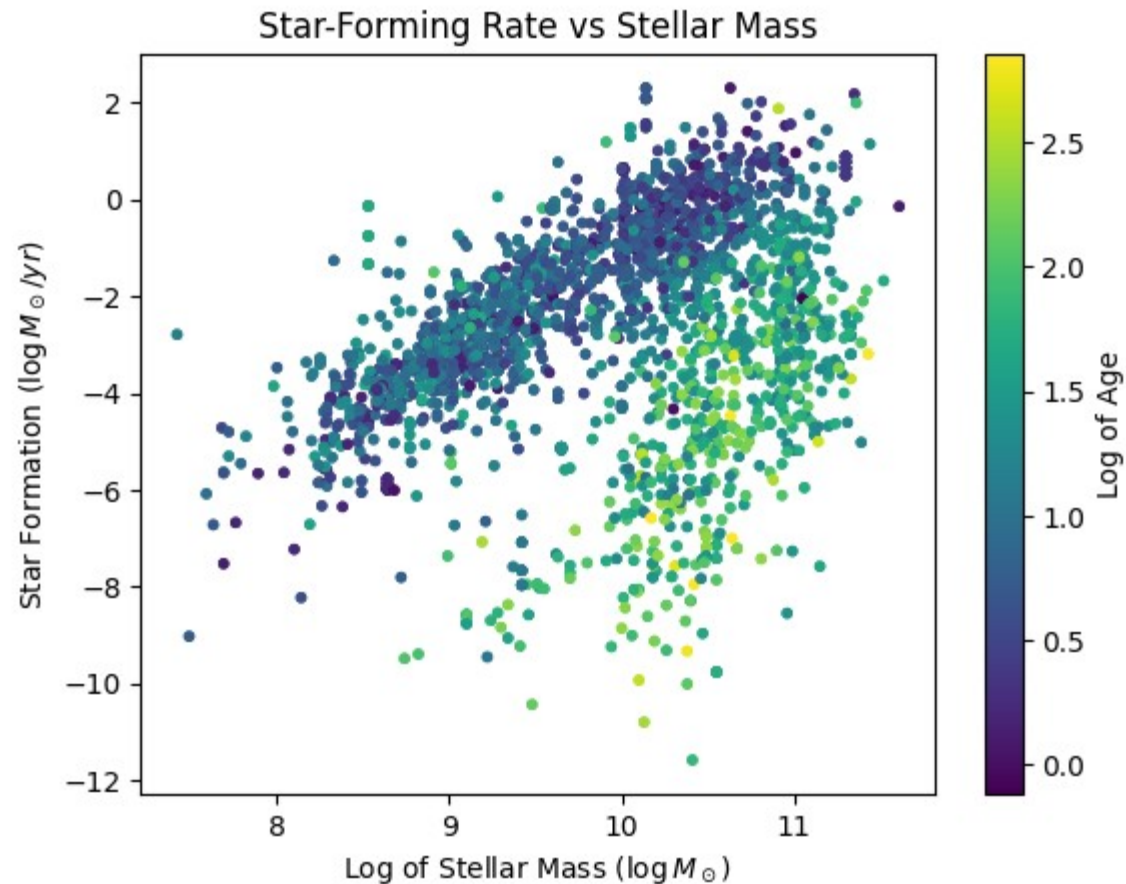
- Star Formation Rate
- Stellar Mass
- I affectionately call this the “stapler” plot.

What can this graph tell us?



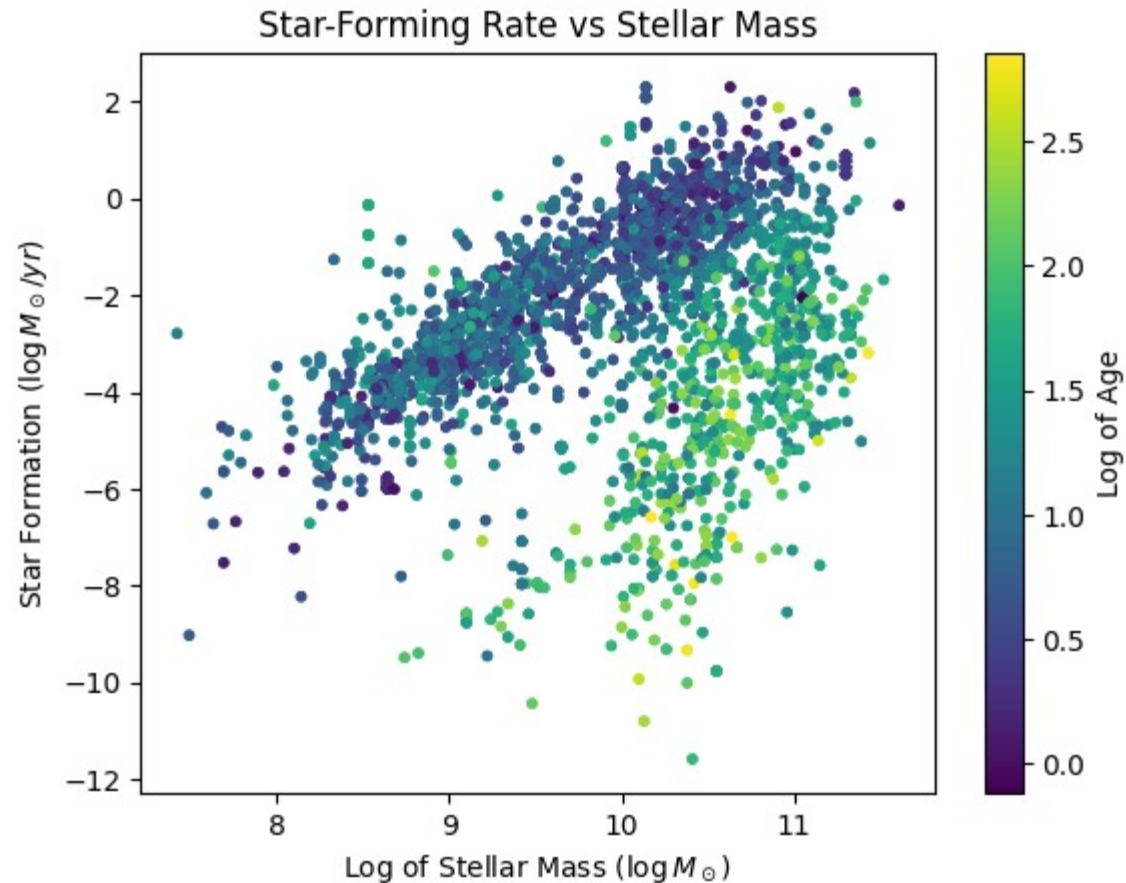
Credit: George et al. 2019

What can this graph tell us?



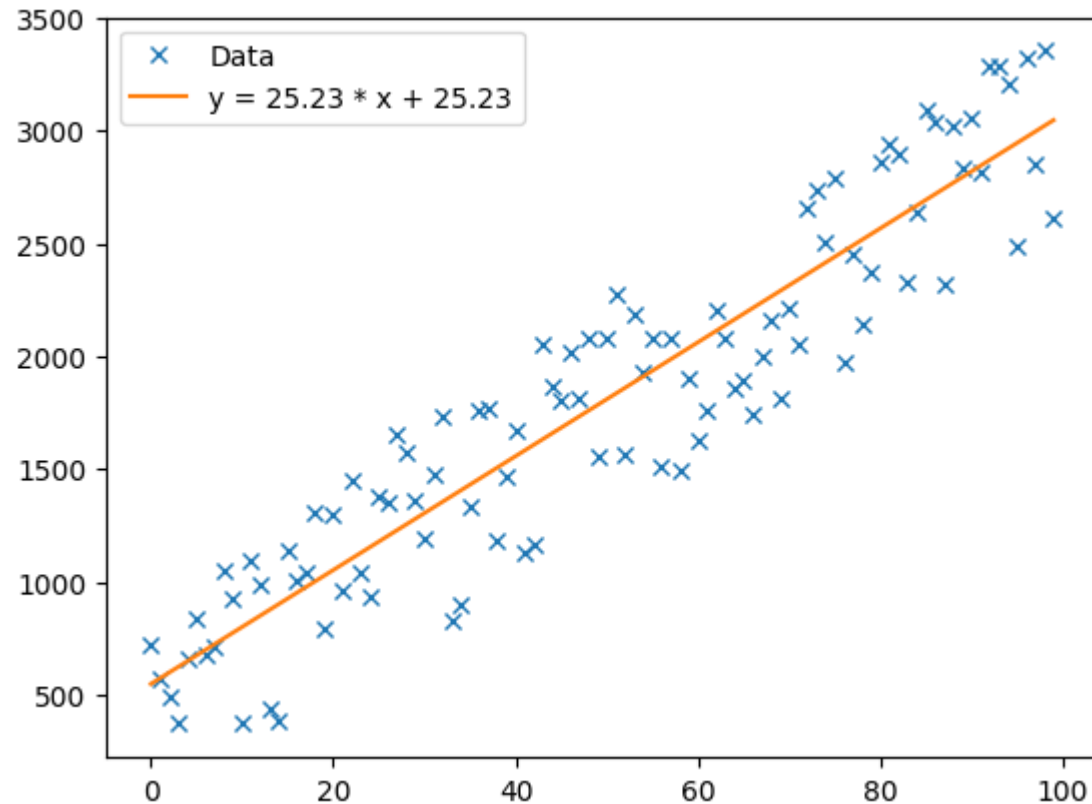
- General trend: Higher mass means higher star-formation rate for star-forming galaxies
- Older galaxies seems to be quenching or quiescent
- Why?

What can this graph tell us?



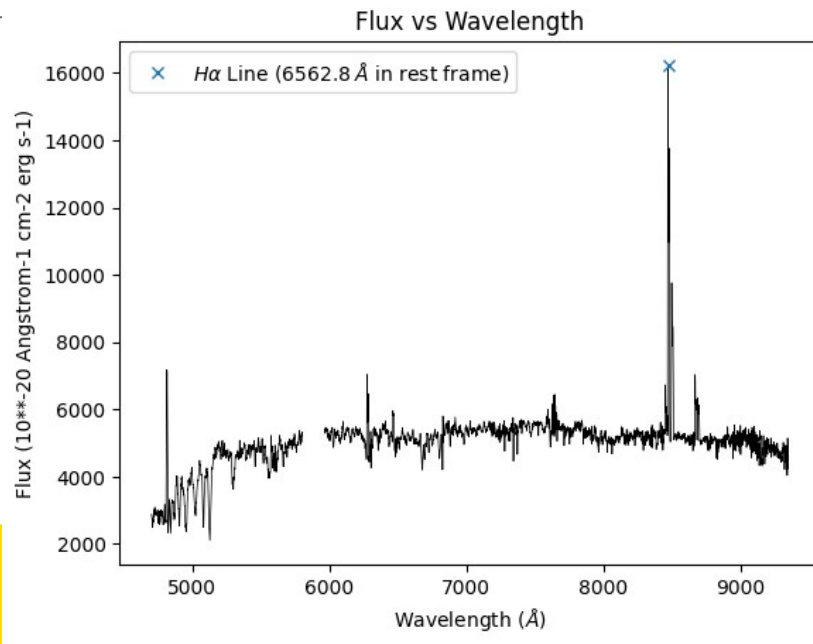
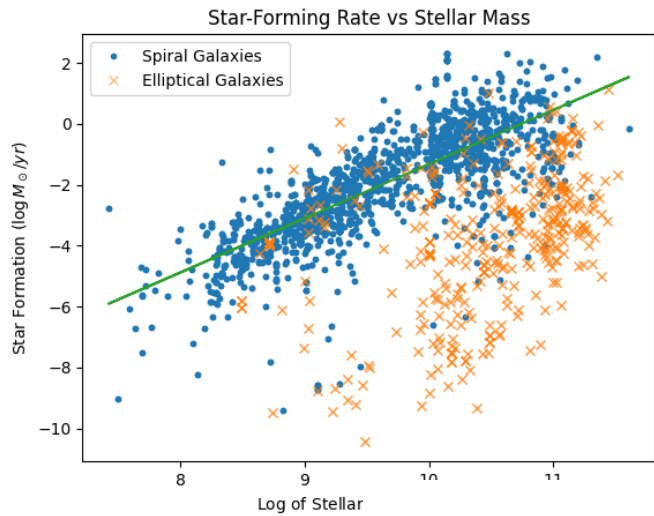
- Greater gravitational potential allows for better gas retention for star-forming galaxies.
- One answer for why older galaxies not being star-forming is used up their gas and don't have the fuel to continue forming stars OR have never be SFMS (need to look more at the chemistry e.g. alpha abundance)
- How long can a galaxy be star-forming? Look into it for PS 1!

Worked Example: Linear Regression



- Lets learn how to perform linear regression with Scipy!

Exercises



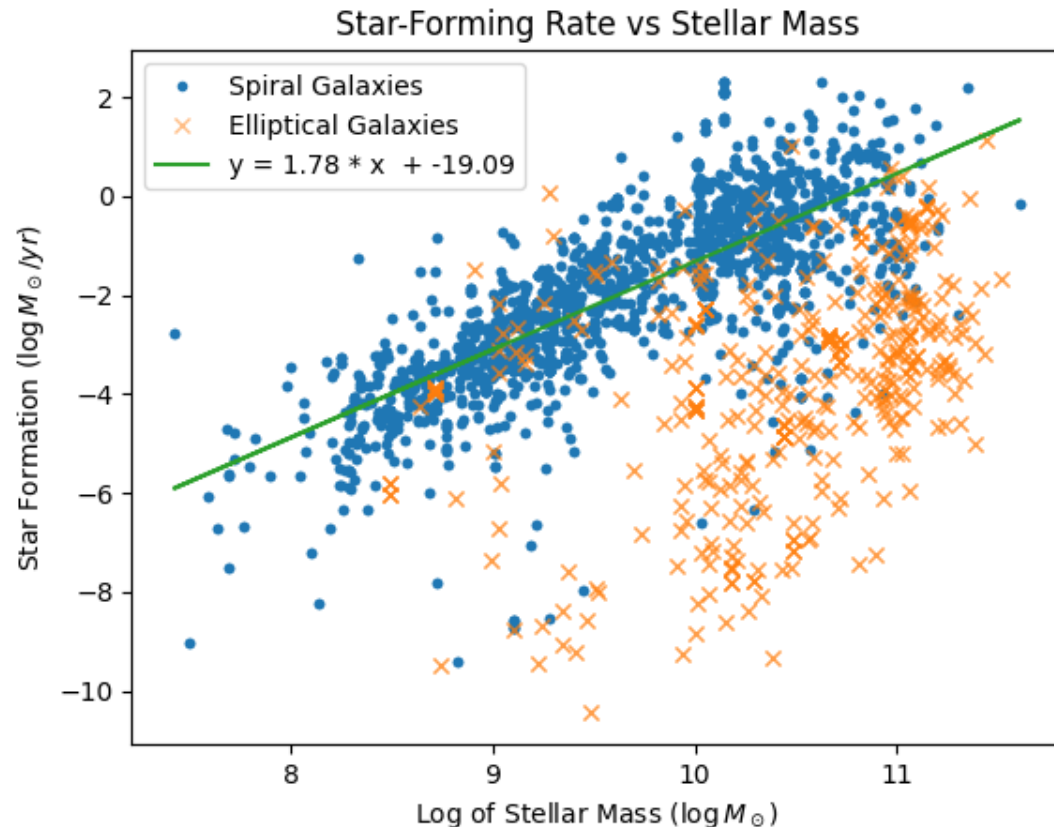
1) Star-Forming Galaxies

- Plot Star Formation Rate vs Stellar mass for spiral and elliptical galaxies. Do these match your expectations?
- Find the star formation rate as a function of stellar mass using linear regression!

2) Spectra of Star-Forming Galaxies

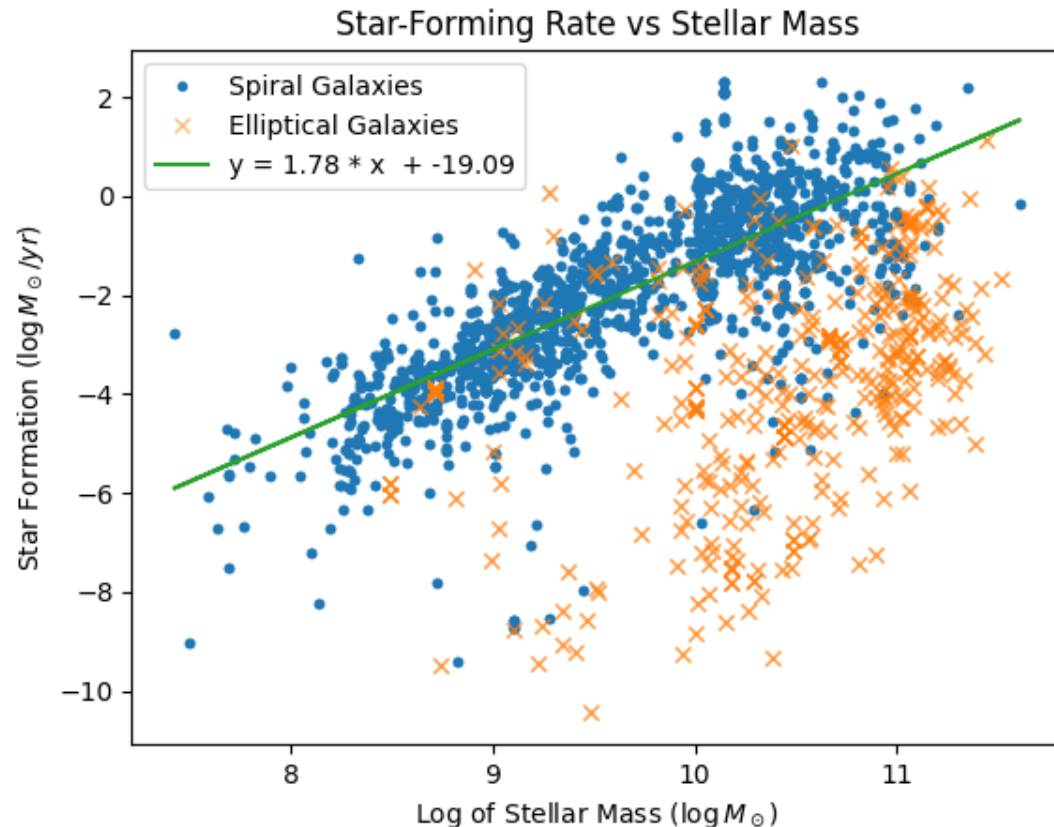
- Compute the red-shift of this particular galaxy given that the Balmer Alpha line is 6562.8 Angstrom in the rest frame. Identify any other relevant peaks
Recall that: $z = \Delta\lambda / \lambda_0$ $\Delta\lambda = \lambda' - \lambda_0$
- Transforming your graph in 2)a) from a redshifted wavelength scale to the rest frame wavelength scale.
Recall that: $\lambda = \lambda' / (1+z)$

Sample Answers



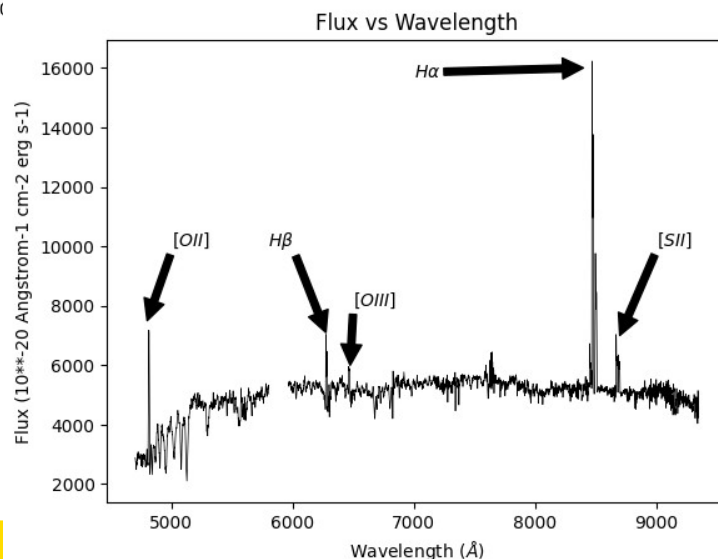
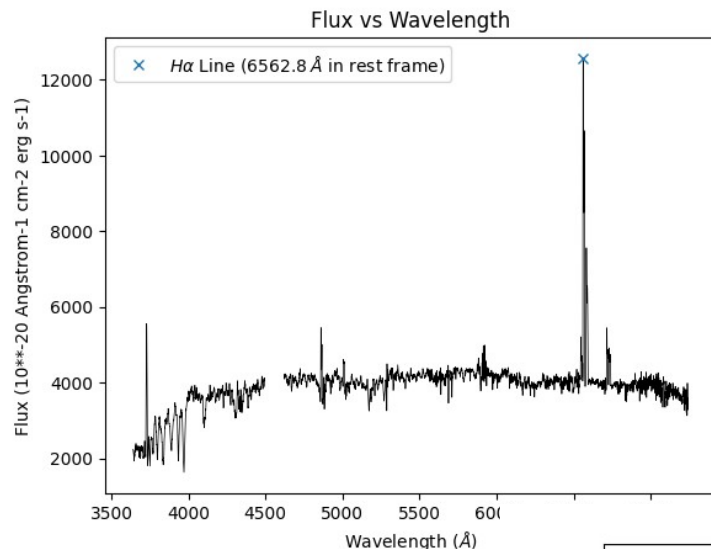
- General Rule of Thumb
 - › Spiral Galaxies tend to be young and star-forming
 - › Elliptical Galaxies tend to be old, red and dead
 - › Variety of environmental factors that could cause a surge or dip in star formation rates
 - Mergers
 - Tidal Interactions
 - AGN Feedback

Sample Answers



- Correlation ~ 0.77
- What causes this variation?
 - } Short process (e.g. dry and wet mergers) (< 2 Gyr)
 - } Long process relating to their evolutionary path
 - AGN Feedback
 - Dark Matter Halo assembly

Sample Answers



- First Image is Flux vs Wavelength in rest-frame (red-shift of ~ 0.29)
- Second image showcases key emission lines
 - } As astronomers, you are expected to be able to identify significant peaks or dips in redshift spectra :)
 - } In practise, what I should've done is fit a Gaussian to the peak but I didn't ;-; (consider doing this if you have free time)